

M-TALHA

SID:- 15882

AI

QW12 no. 01

(H.S)

Qno. 01

Ans. 01 An agent is anything that can perceive its environment & act on it.

Core Role Bridge between environment & intelligence behaviour.

AI Agents Interact with its Environment:-

The agents get inputs (percepts) from its environment & takes ^{action} decision that affects _{on} it.

Cycles:-

Perception → Decision → Action → Environment

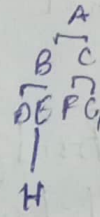
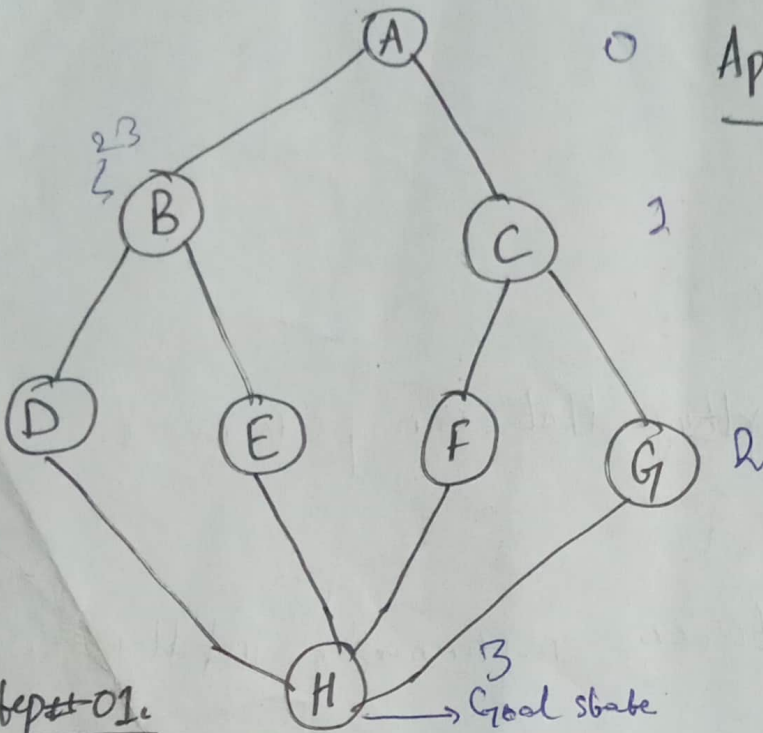
1. Perception → receive data from the environment
2. Decisions → uses a rule set, ML model or reinforcement learning policy to choose an action
3. Action → Performs an action (move or robot)

Diagram 2

(H.S)

Q no. 02

Applying BFS



A-H

Step#01.

[A]

[B | G | H]

[B | C]

[C | H]

[D | E]

[H]

[D | E | F | G]

[E | F | G | H]

~~H → B → C → D → H~~

Path:

[A → B → C → D → E → F → G → H]

Find shortest Path → ~~A → B → D → H~~

Time Complexity:

$$T.M = O(V+E) \Rightarrow O(6^4) = O(2^8) = 8$$

nodes $\Rightarrow V = 8$

Edges $\Rightarrow E = 10$